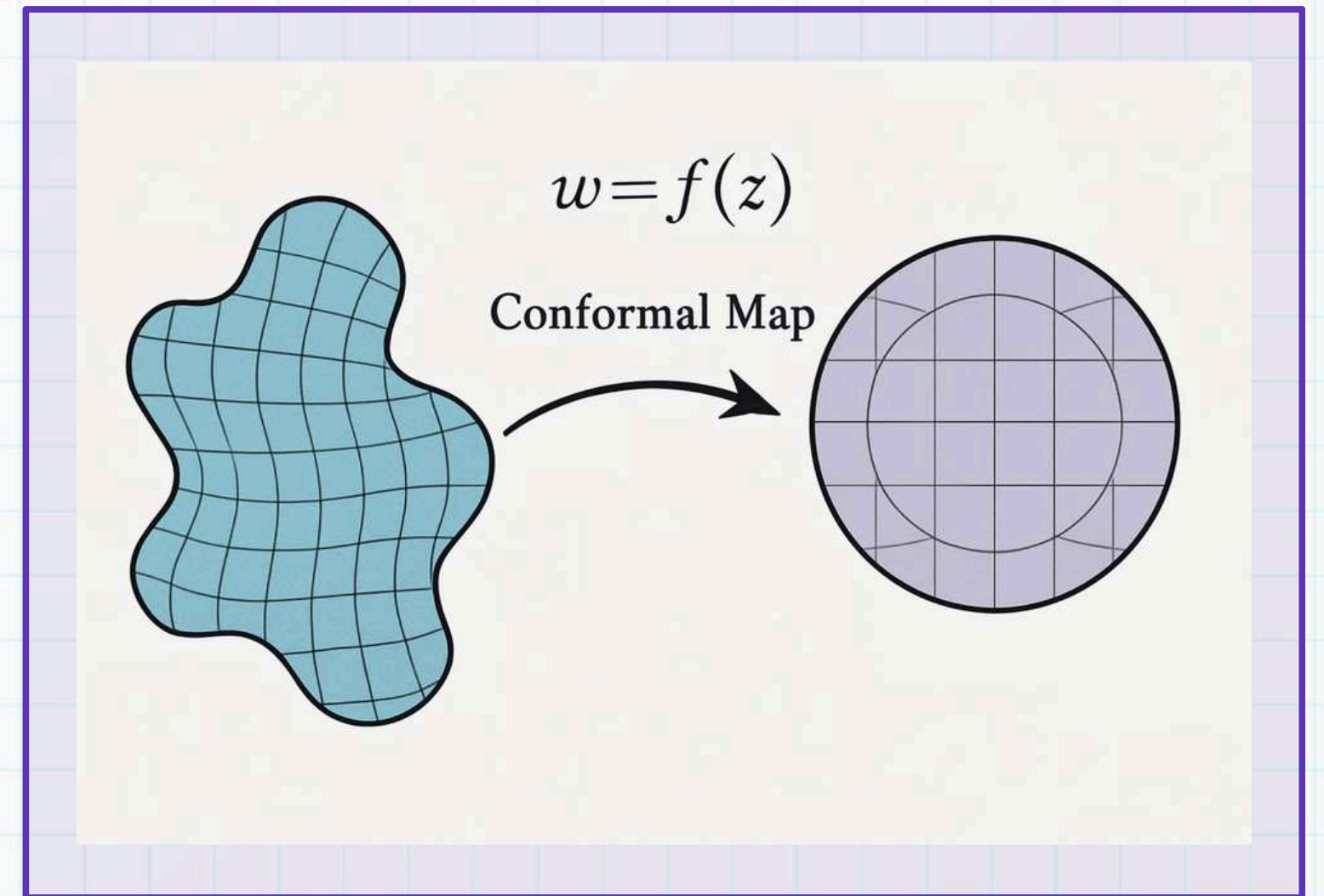


Computation of Conformal Mappings

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Exploring the theory and practice of conformal transformations in complex analysis, with applications to engineering and physics.

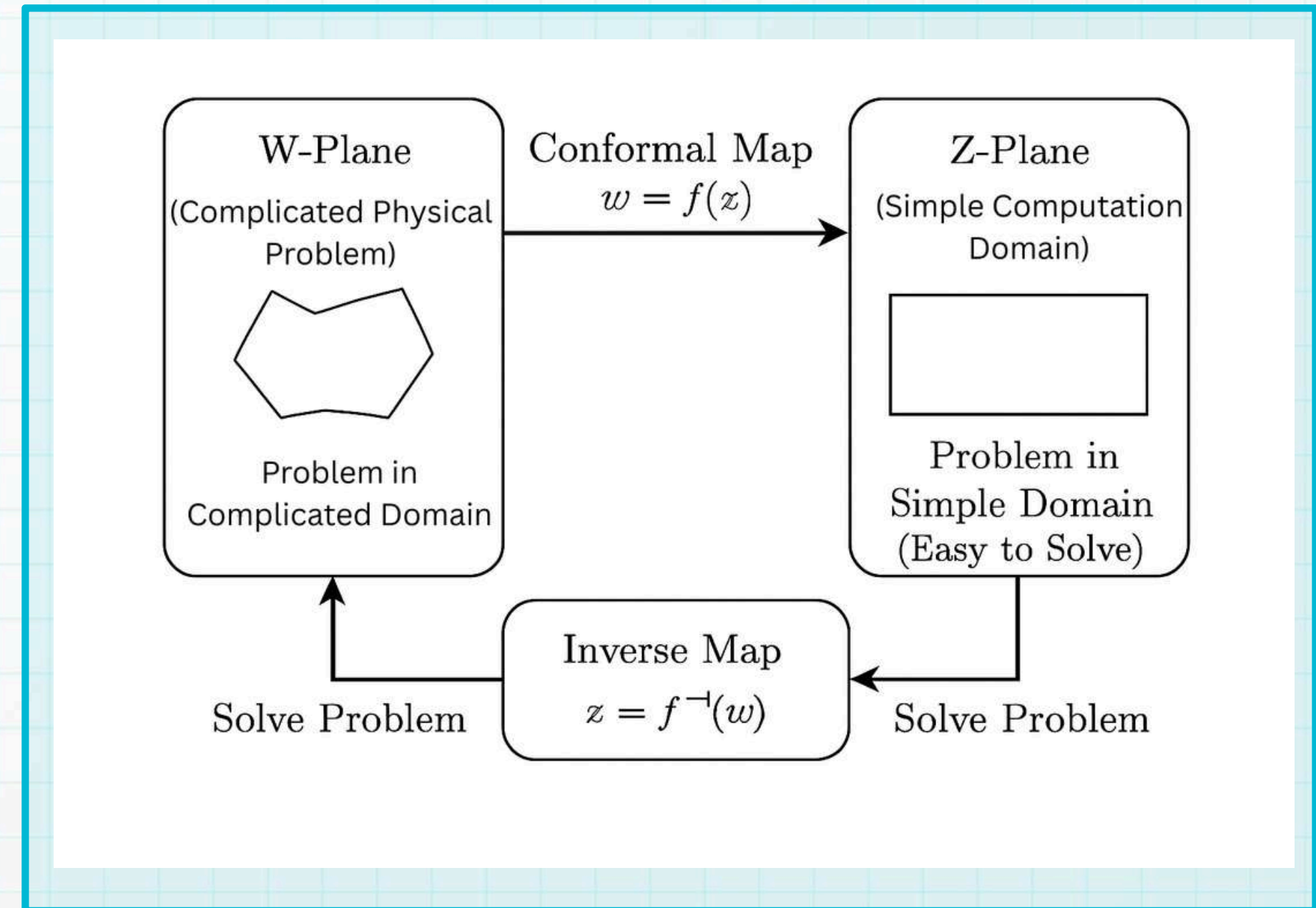
Why Conformal Mapping?

Transforming complex geometries into solvable problems

The Challenge: Physical problems require solutions in irregular domains, but analytical methods work only in standard geometries

The Solution: Conformal mapping transforms complex boundaries into simple domains while preserving governing equations

The Impact: Enables closed-form analytical solutions for problems that would otherwise require intensive numerical computation



The Riemann Mapping Theorem

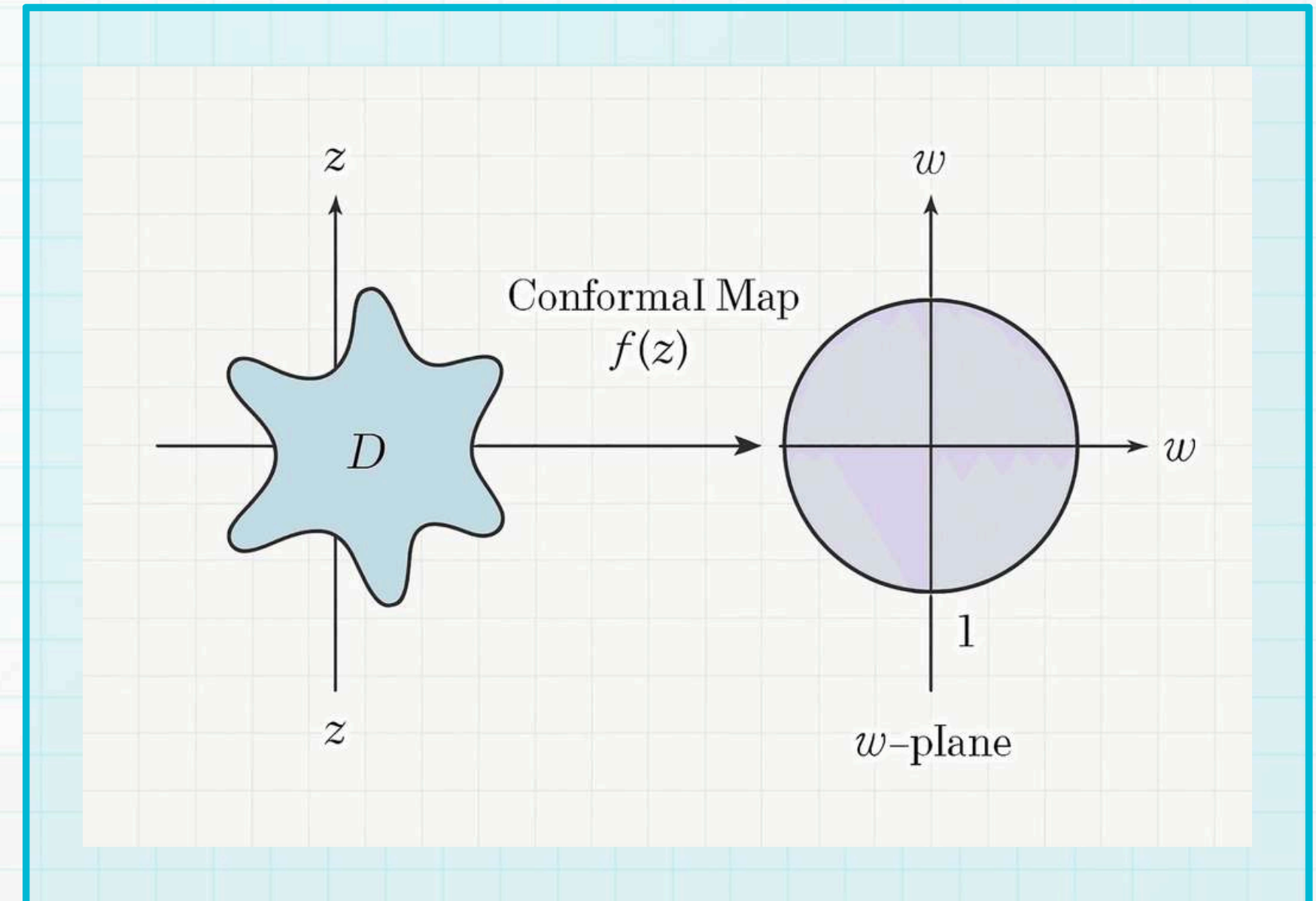
Theorem: Let $D \subset \mathbb{C}$ be a simply connected region. Then there exists a conformal mapping $f: D \rightarrow U$ that maps D onto the open unit disk $U = \{w \in \mathbb{C} : |w| < 1\}$.

Existence: The theorem guarantees that a conformal mapping from any simply connected domain to the unit disk always exists.

Uniqueness: The mapping is unique up to normalization (e.g., fixing the image of a point and the argument of the derivative).

Significance: This is the foundational result that enables the reduction of complex boundary value problems to standard domains.

Historical Impact: Proved by Riemann in 1851; rigorously established by Schwarz (1890) and Hilbert (1901).



Foundations of Complex Analysis

Analytic Function

A function $f(z)$ is analytic (holomorphic) in region D if its derivative $f'(z)$ exists at every point in D .

Cauchy-Riemann Equations

Necessary and sufficient conditions for analyticity: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$
and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

Harmonic Functions

The real and imaginary parts of an analytic function are harmonic:
 $\nabla^2 v = 0$ and $\nabla^2 u = 0$ (Laplace's equation).

Connection to Conformal Mappings

Analyticity ensures angle preservation and local one-to-one mapping, the core properties of conformal transformations.

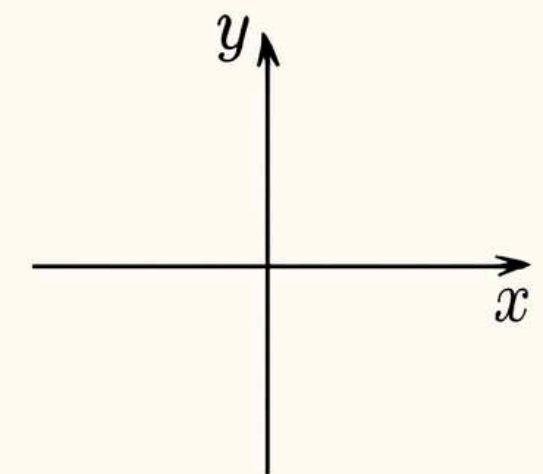
Real part

Imaginary part

$$w = f(z) = u(x, y) + iv(x, y)$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$



Conformal Mapping: Definition and Conditions

Definition:

A transformation $w=f(z)$ is conformal at point z_0 if it preserves the angle between any two curves passing through z_0 , both in magnitude and sense (orientation).

Sufficient Condition:

If $f(z)$ is analytic in domain D and $f'(z) \neq 0$ for all $z \in D$, then $f(z)$ is a conformal mapping in D .

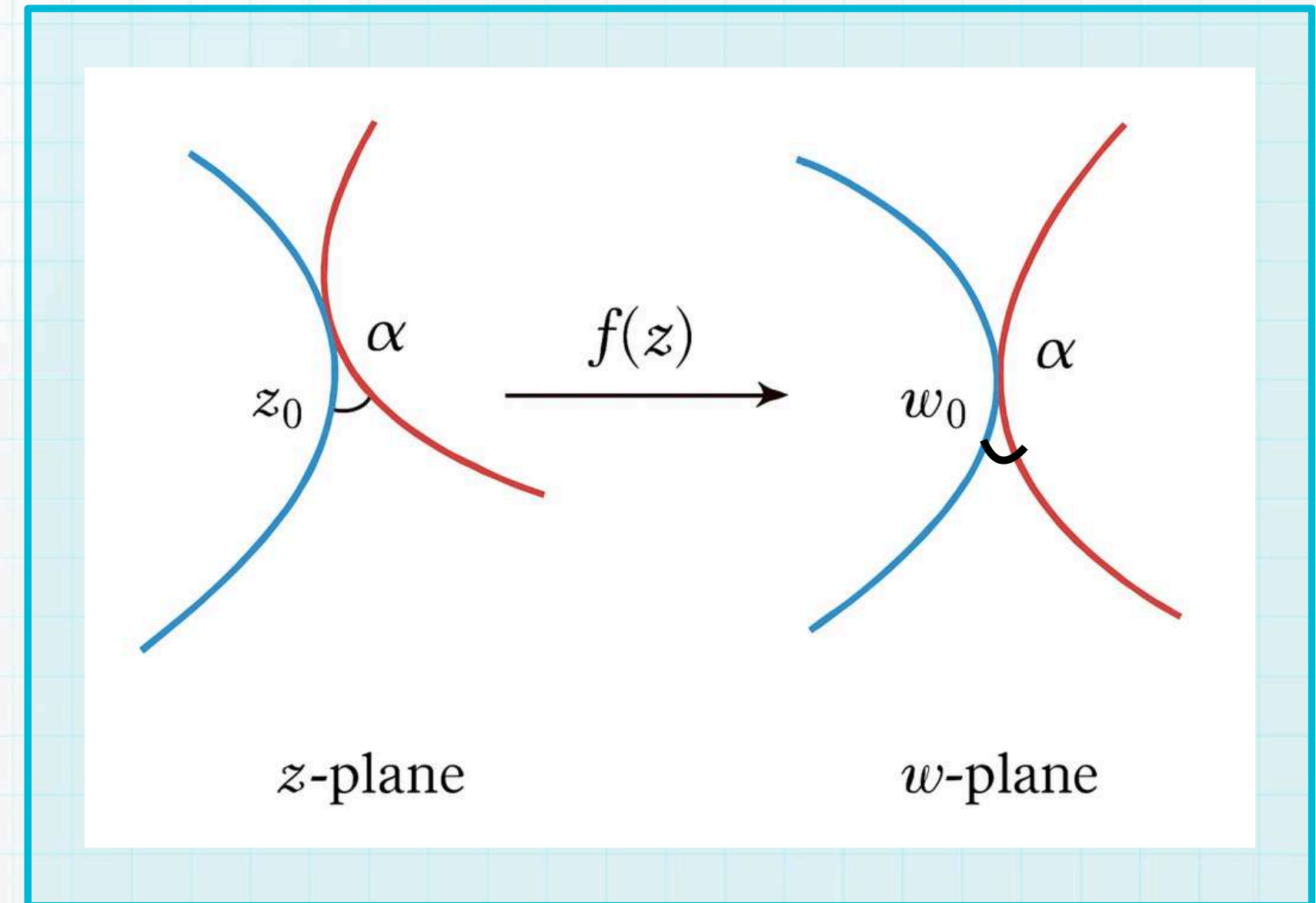
Local Univalence:

The condition $f'(z) \neq 0$ ensures the mapping is locally one-to-one (injective), meaning small neighborhoods map to small neighborhoods without overlap.

Key Property:

Conformal mappings preserve the orthogonality of curves.

If two curves intersect at right angles in the z -plane, their images also intersect at right angles in the w -plane.



Bilinear (Möbius) Transformation

$$w = (az + b) / (cz + d)$$

where $ad - bc \neq 0$

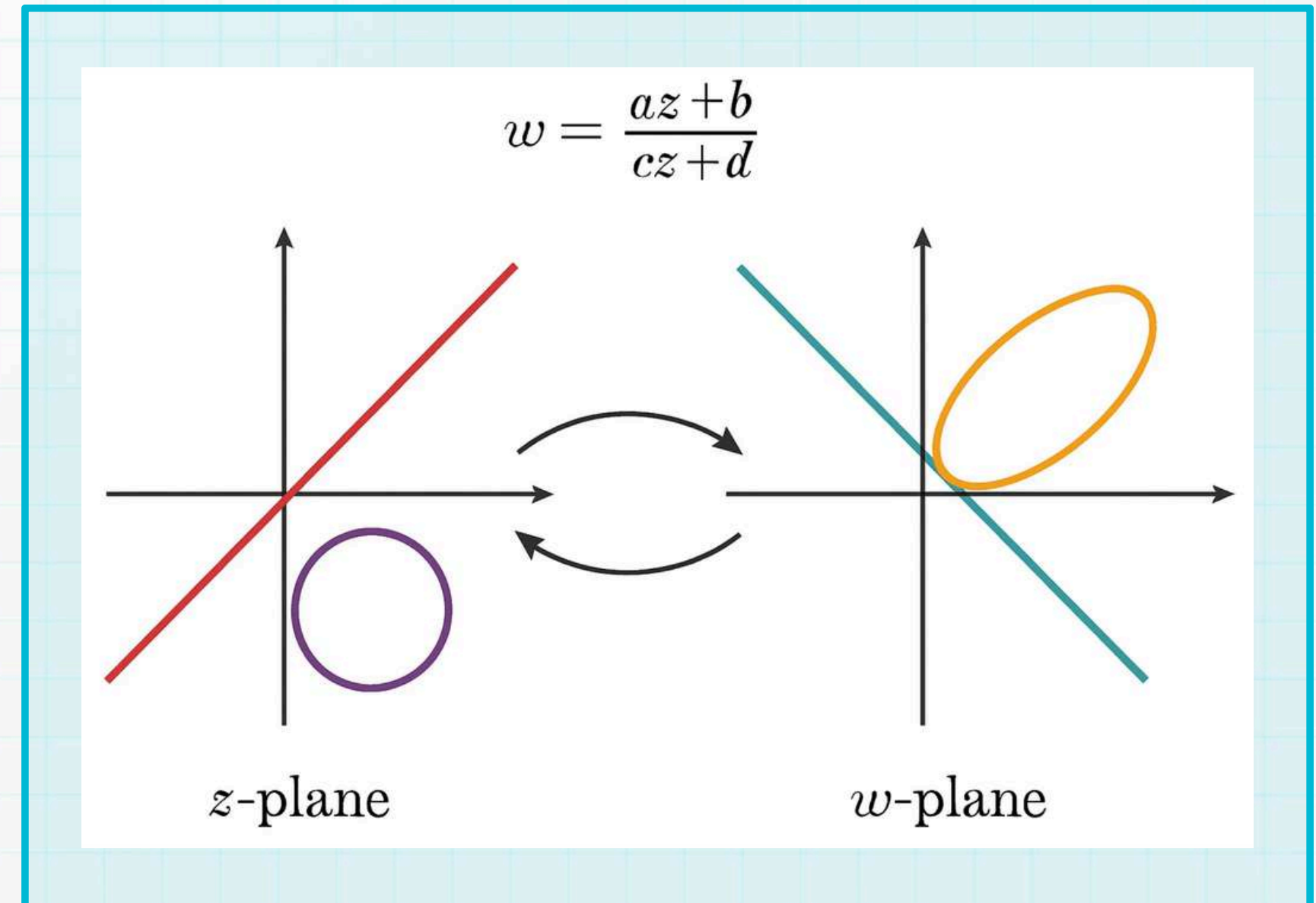
Conformality: The transformation is conformal everywhere except at the pole $z = -d/c$, where the derivative is undefined.

Circle/Line Preservation: Maps circles and straight lines in the z -plane to circles or straight lines in the w -plane. Lines are treated as circles with infinite radius.

Composition: Any bilinear transformation can be decomposed into a sequence of simpler transformations: translation, rotation/magnification, and inversion.

Normalization: The condition $ad - bc \neq 0$ ensures the transformation is non-degenerate and invertible.

Application: Commonly used to map the upper half-plane to the unit disk, or to transform one standard region to another.



Schwarz-Christoffel Transformation

Purpose: Maps the upper half-plane ($\Im(z) > 0$) conformally onto the interior of any simple polygon with finitely many vertices.

$$dw/dz = A \prod_{k=1}^n (z - x_k)^{-\alpha_k}$$

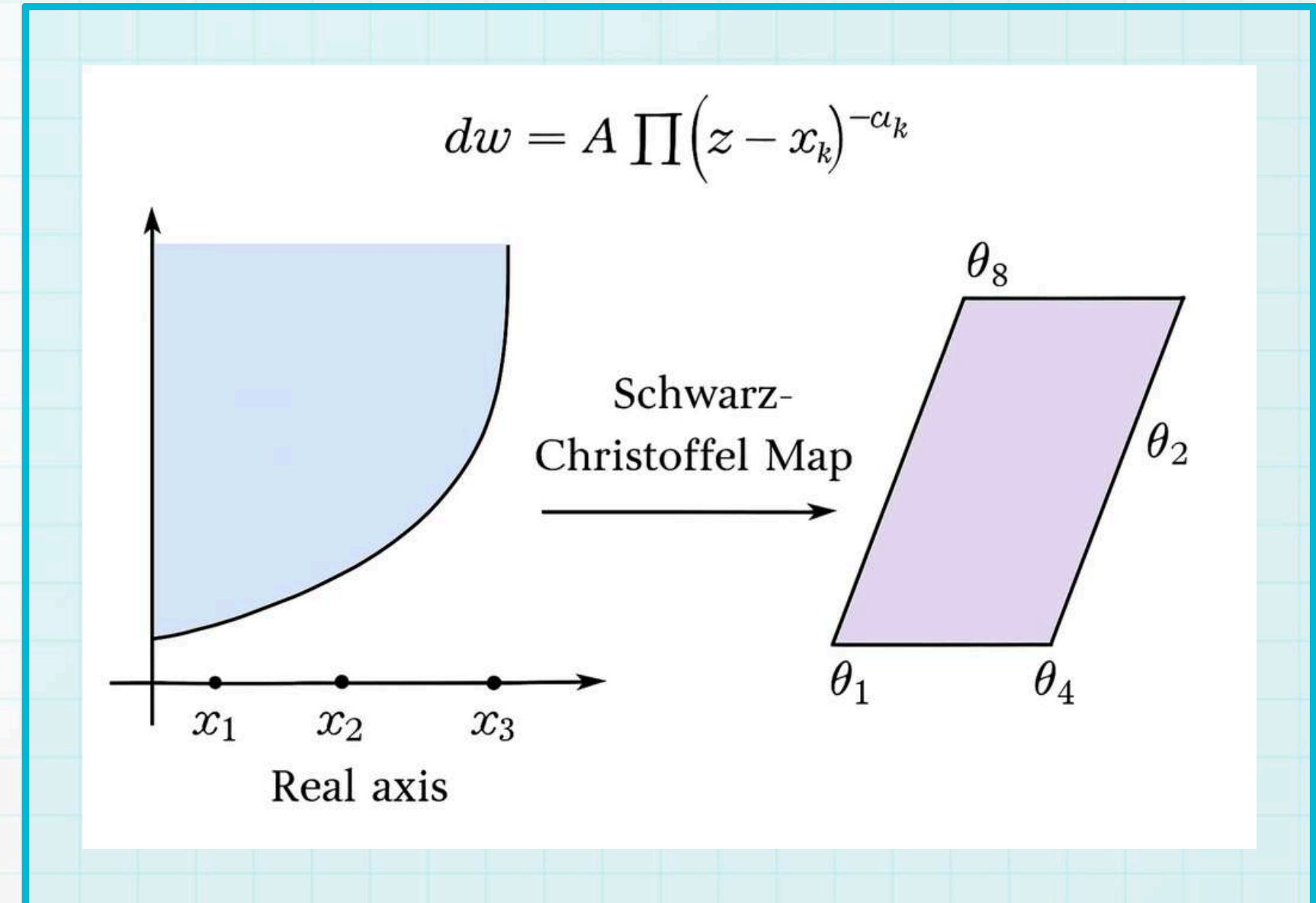
$$w = \int dw/dz \, dz + B$$

Parameters:

- x_k : Points on real axis mapping to polygon vertices
- α_k : Exterior angles (related to interior angles $\theta_k = (1 - \alpha_k)\pi$)
- A, B : Complex constants for normalization

Key Property: The real axis in the z -plane maps to the boundary of the polygon in the w -plane, with vertices occurring at $x_1, x_2, \dots, x_n$.

Significance: Provides an explicit formula for conformal mapping of polygonal regions, solving a classical problem in complex analysis.



Schwarz-Christoffel: Derivation Concept

The Change in Argument

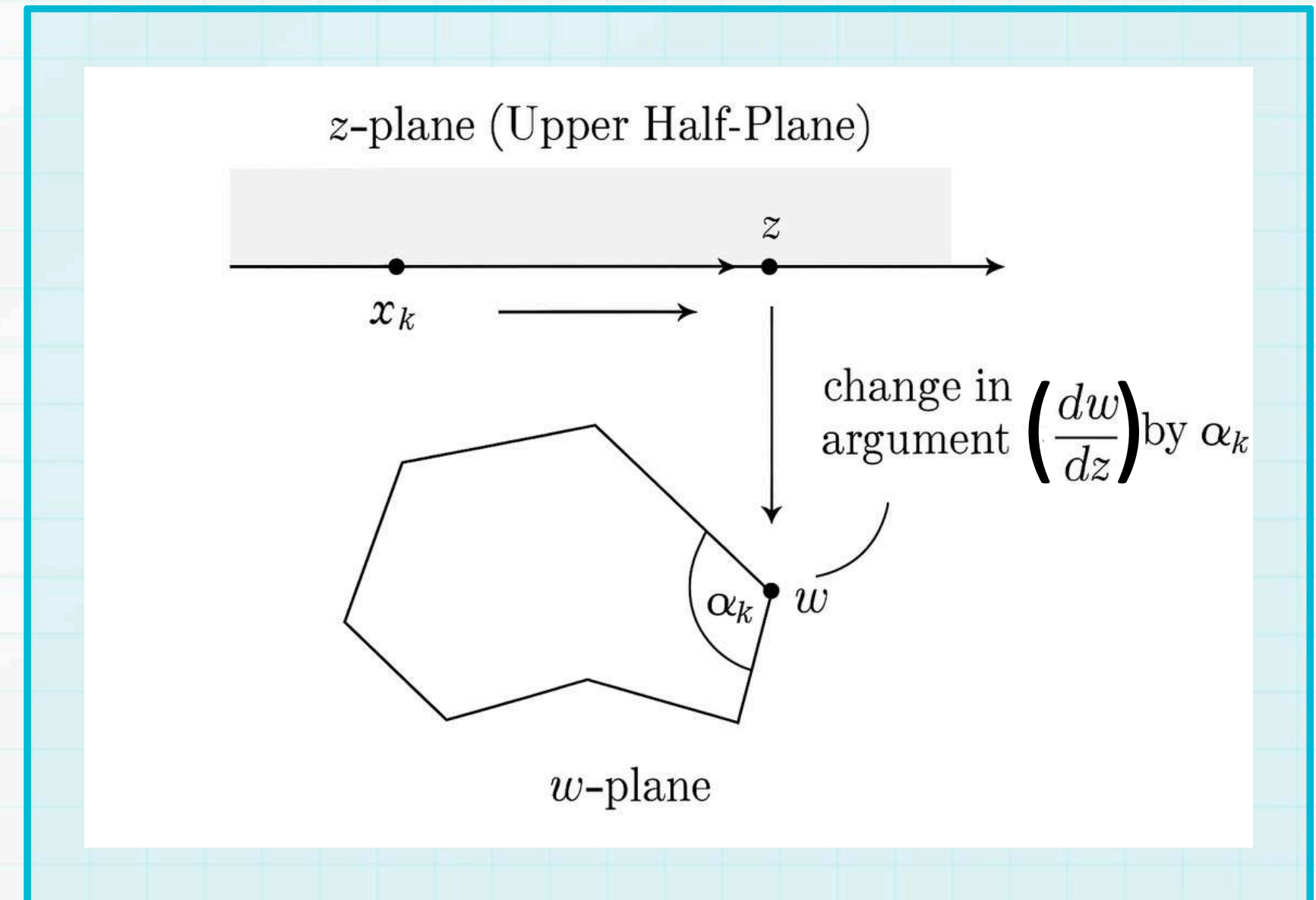
The core idea is to control the change in the argument (angle) of dw/dz as z moves along the real axis (the boundary of the upper half-plane).

Boundary Mapping

The real axis in the z -plane maps to the boundary of the polygon in the w -plane. As z traverses the real axis from left to right, w traces the polygon boundary.

Vertex Control

At each vertex w_k of the polygon, the path must turn by the exterior angle $\alpha_k \pi$. This turn is achieved by controlling the argument change of dw/dz .



Exterior Angle Relationship

Interior angle: $\theta_k = (1 - \alpha_k)\pi$

Exterior angle: $\alpha_k \pi$

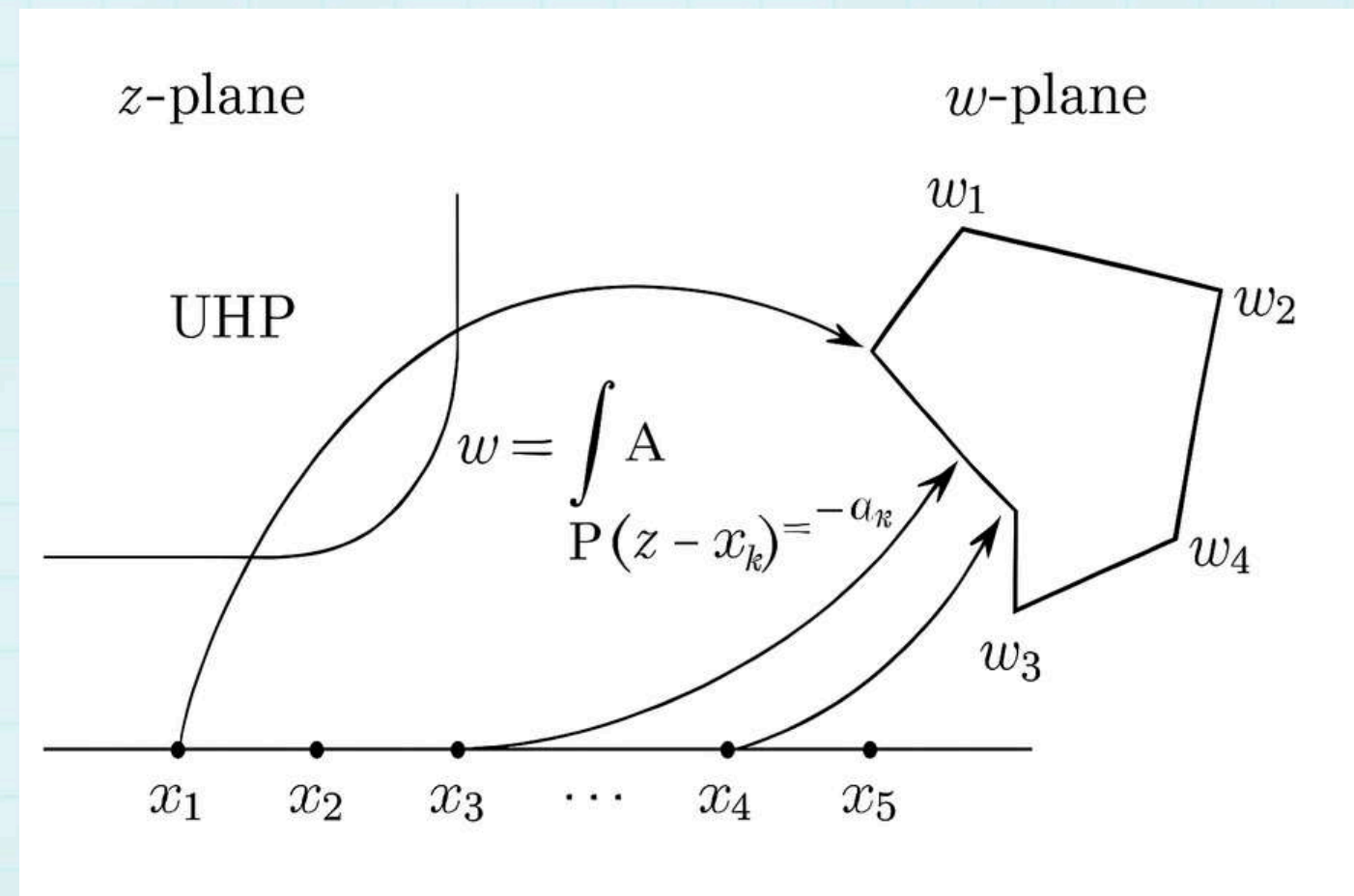
Schwarz-Christoffel: Construction Steps

Schwarz-Christoffel Formula

$$dw/dz = A \prod (z - x_k)^{-\alpha_k}$$

Why $(z - x_k)^{-\alpha_k}$?

As z crosses x_k from left to right, the argument of $(z - x_k)^{-\alpha_k}$ changes by exactly $-\alpha_k \pi$, producing the required turn at vertex w_k .



Step 1: Identify vertices $x_1, x_2, \dots, x_n$ on the real axis that map to polygon vertices $w_1, w_2, \dots, w_n$.

Step 2: Determine exterior angles $\alpha_k \pi$ from the interior angles θ_k of the polygon.

Step 3: Construct the product $\prod (z - x_k)^{-\alpha_k}$ to ensure each factor contributes the correct argument change.

Step 4: Integrate to obtain $w = A \int \prod (z - x_k)^{-\alpha_k} dz + B$, where A and B are determined by normalization conditions.

Schwarz-Christoffel Example: Rectangle Mapping

$$\frac{dw}{dz} = A \prod_{k=1}^4 (z - x_k)^{-1/2} = \frac{A}{\sqrt{(z - x_1)(z - x_2)(z - x_3)(z - x_4)}}$$
$$w = A \int \frac{dz}{\sqrt{(z - x_1)(z - x_2)(z - x_3)(z - x_4)}} + B$$

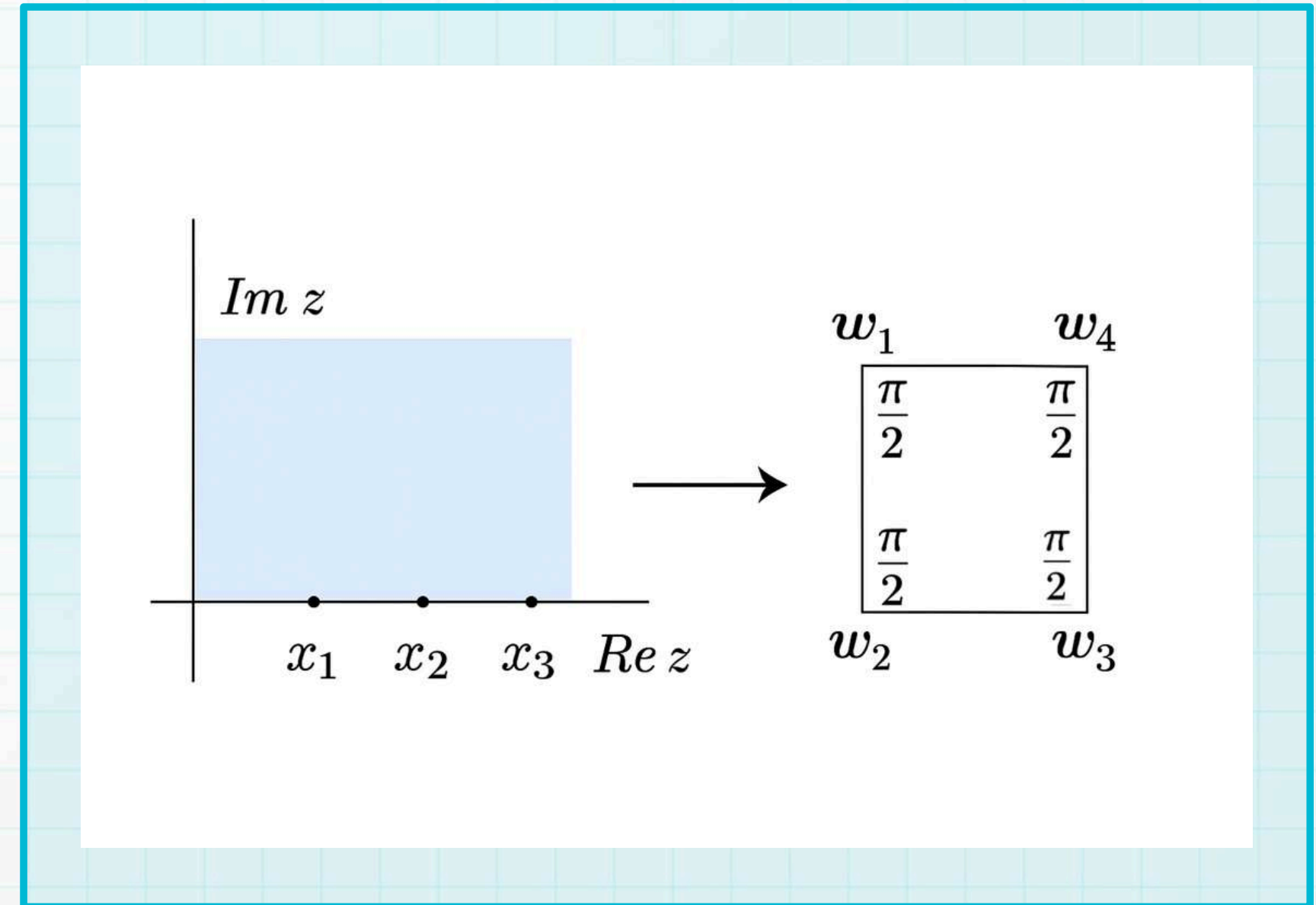
Target Domain: Map the upper half-plane (UHP) onto a rectangle in the w -plane.

Vertices & Angles: A rectangle has four vertices with interior angles $\pi/2$ at each corner. Exterior angles: $\alpha_k = 1 - (\pi/2)/\pi = 1/2$.

Boundary Points: Points x_1, x_2, x_3, x_4 on the real axis map to vertices w_1, w_2, w_3, w_4 of the rectangle.

Integral Form: The mapping involves an elliptic integral, demonstrating the complexity of even simple polygonal mappings.

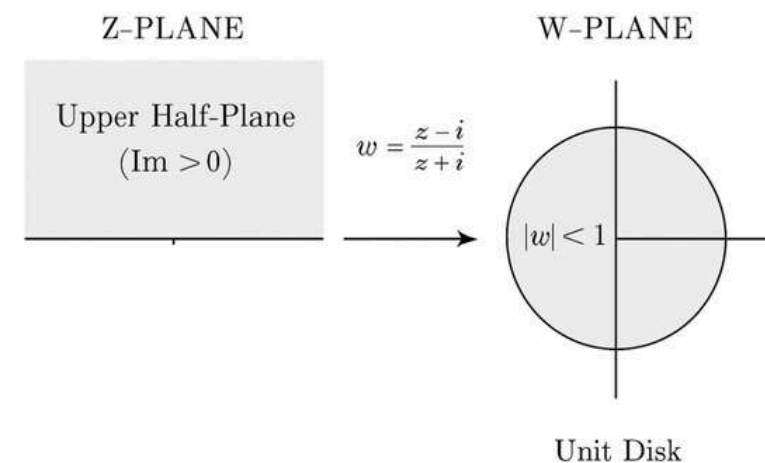
Note: The constants A and B are determined by normalization conditions (e.g., fixing the image of specific points).



Schwarz-Christoffel: More Examples

Example 1: Unit Disk

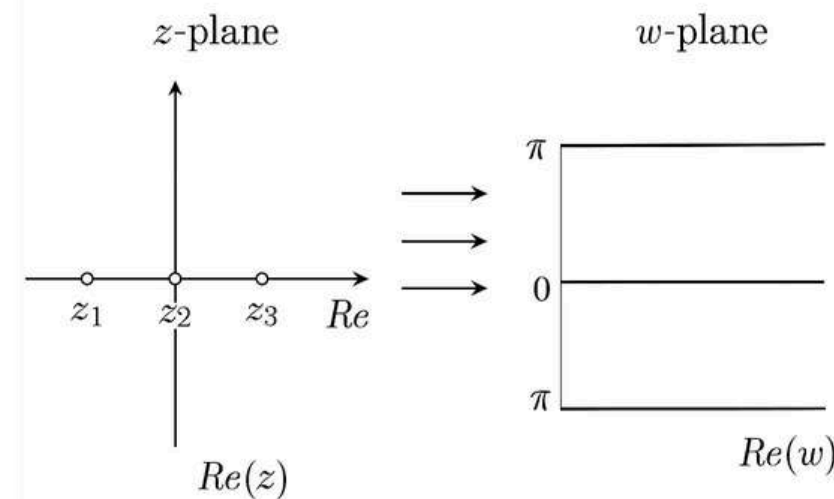
$$w = \frac{z - i}{z + i}$$



Boundary Points: All points on the real axis
Exterior Angle: N/A (Conformal mapping of a simply connected domain).
Application: Riemann Mapping Theorem, Potential Theory, Bilinear Transformation.

Example 2: Semi-infinite Strip

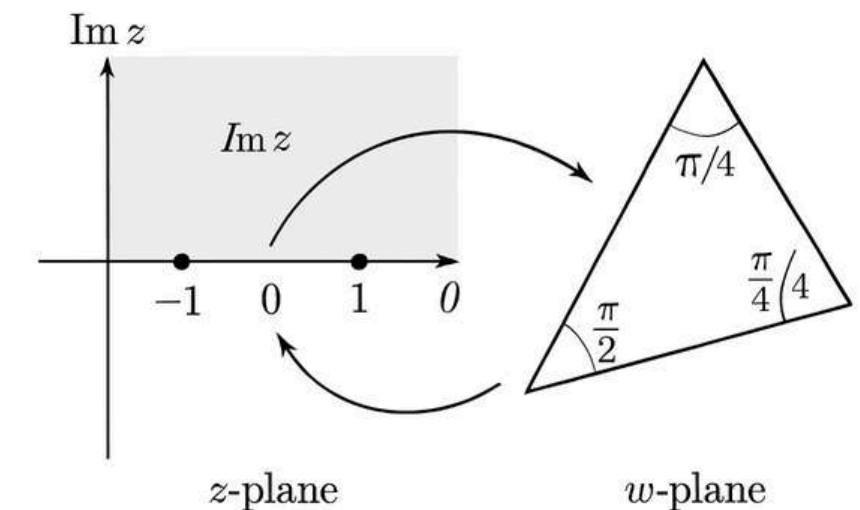
$$\frac{dw}{dz} = A(z^{-1/2})(z - z_1)^{-1/2}(z - z_2)^{-1/2}$$



Boundary Points: Three points on real axis
Exterior Angles: All $1/2$
Application: Channel flows, waveguides

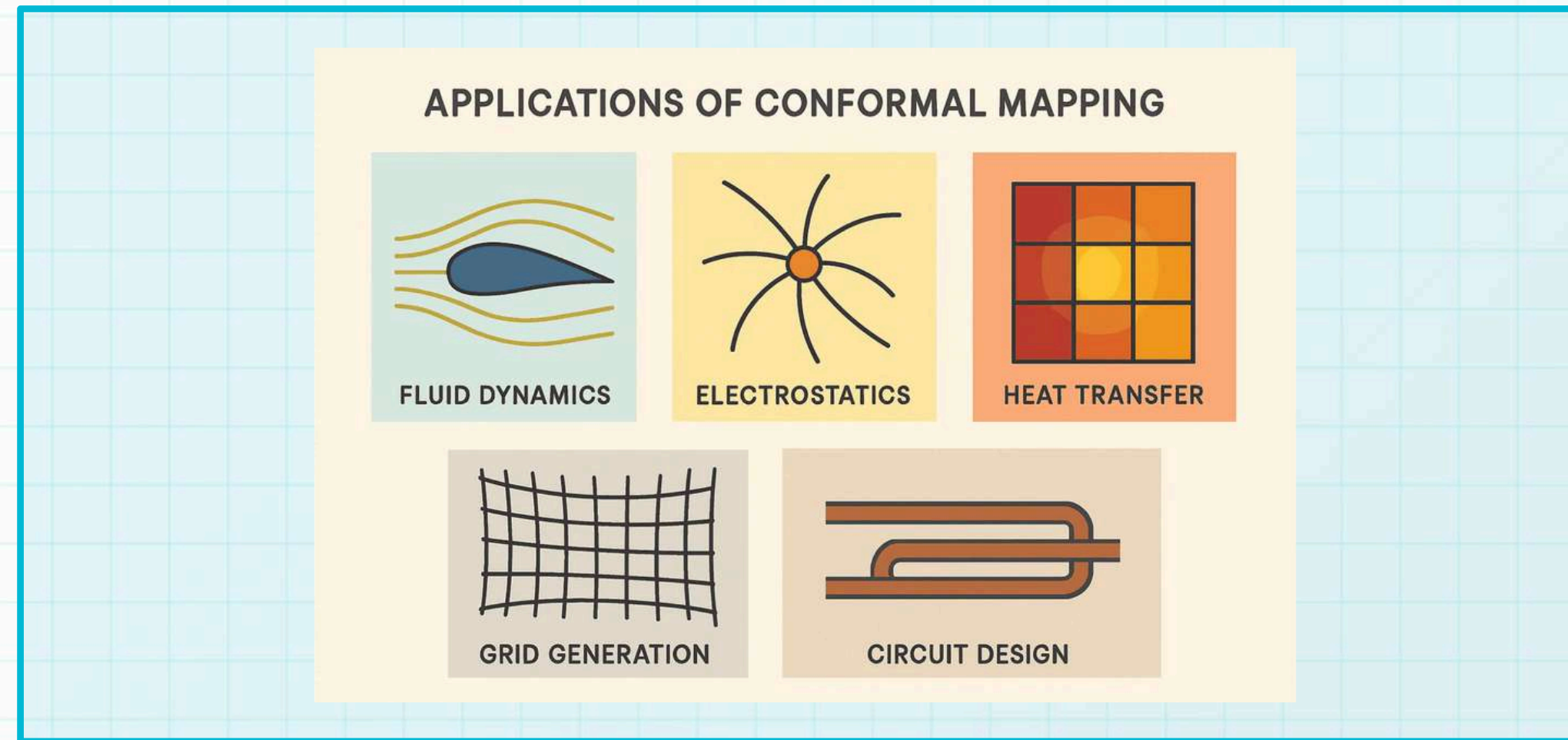
Example 3: Triangular Region

$$\frac{dw}{dz} = A z^{-1/2} (z + 1)^{-1/2} (z - 1)^{-1/2}$$



Boundary Points: $x_1 = -1, x_2 = 0, x_3 = 1$
Interior Angles: $\pi/4, \pi/4, \pi/2$
Application: Elastic membranes, fluid flow

Real-World Applications of Conformal Mapping



Fluid Dynamics

Analyzing potential flow around airfoils and complex geometries. The Joukowski transformation maps a circle to an airfoil shape, enabling the calculation of lift and drag forces in aerodynamics.

Electrostatics

Solving for electric potential and field lines in complex geometries. Conformal mapping transforms irregular conductor configurations into standard domains where solutions are known.

Heat Transfer

Analyzing steady-state heat conduction in two-dimensional domains with irregular boundaries. Conformal mapping reduces the problem to a standard domain with simpler boundary conditions.

Conclusion and Future Directions

Theoretical Foundation

The **Riemann Mapping Theorem** guarantees existence and uniqueness of conformal mappings from any simply connected domain to the unit disk, providing the theoretical basis for all computational methods.

Key Transformations

Schwarz-Christoffel formula provides explicit mappings to polygonal regions, while **Bilinear transformations** offer simple, explicit solutions for circular and half-plane domains.

Conformal Mapping

Angle-preserving transformations defined by analytic functions with $f'(z) \neq 0$, enabling reduction of complex boundary value problems to standard domains.

Future Scope

- Validation through simulations/examples
- Numerical approximation of mappings
- Advanced conformal mapping techniques
- Boundary-value problem applications

Thank You

Questions & Discussion

Conformal mapping remains a powerful and elegant tool in mathematics, with applications spanning engineering, physics, and computational science. The interplay between theory and computation continues to drive innovation in this field.